

CO-2 AT-1
Analytical Problem Solving

S.A. Mohammed
Fareeth
192511288
CSA0307

1) Convert infix expression to postfix

$$B - (C/D + (E \% F * G) / H) * J$$

Ans

Input	Stack	Output
B	Empty	B
-	-	B
(	-(	B
C	-(	BC
/	-(/	BC/
D	-(/	BCD
+	-(+	BCD/
(	-(+(	BCD/
E	-(+(	BCD/E
%	-(+(%	BCD/E%
F	-(+(%	BCD/EF
*	-(+(*	BCD/EF%
G	-(+(*	BCD/EF%.G
)	-(+)	BCD/EF%.G*
/	-(+/	BCD/EF%.G*/
H	-(+/	BCD/EF%.G*/H
)	-	BCD/EF%.G*/H/
*	-*	BCD/EF%.G*/H/+
J	-*	BCD/EF%.G*/H/+/J
End	Empty	BCD/EF%.G*/H/+/J*

Evaluate Postfix

a) 9 3 8 4 / * +

Symbol	Stack
9	9
3	9 3
8	9 3 8
4	9 3 8 4
/	9 3 2
*	9 6
+	15

Answer = 15

b) 6 2 + 6 3 2

Impossible because not enough operators to evaluate all operands

6 2 + $\rightarrow$ 8
remaining = 6 3 2

2) Queue Operations

Initial Queue : Front = -1
Rear = -1
Queue = []

Operations	Queue	Front	Rear
Enqueue (2)	2	0	0
Enqueue (67)	2, 67	0	1
Enqueue (89)	2, 67, 89	0	2
Dequeue ()	67, 89	1	2

Enqueue (15)	67, 89, 15	1	3
Enqueue (30)	67, 89, 15, 30	1	4
Enqueue (52)	Queue overflow	1	4
Dequeue ()	89, 15, 30	2	4
Dequeue ()	15, 30	3	4
Dequeue ()	30	4	4
Dequeue ()	Empty Queue	Front = -1	Rear = -1

Final Result :

Overflow

Occurs during:

Enqueue (52)

Elements Removed: 21, 67, 89, 15, 30

Final Queue: Queue = []

Front = -1

Rear = -1